

$AnAn$ versus PL($AnAn$) photon fluxes: why our plot and arXiv:2606.05469 look different

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1 Summary

In `python/flux_comparison.py` our $AnAn$ flux crosses *above* the point-like PL($AnAn$) flux at $z \gtrsim 0.03$, whereas in Fig. `flux_AA_classes` of Guzey, Innocenti, Staśto and Strikman (arXiv:2606.05469) the PL flux is *always* the larger one. The two plots do not contradict each other: the curves labelled $AnAn$ are different quantities.

- Their $AnAn$ flux counts every photon at the centre of the target nucleus (single impact parameter b , weighted by $\Gamma_{AA}(b)$).
- Our $AnAn$ curve is the *effective flux* of Eskola et al. (arXiv:2404.09731, Eq. 4), which averages over the positions of the target nucleons.

Evaluated with the same single- b definition as theirs, our code also gives a flux below PL($AnAn$) at every z .

2 Why the smallest distance controls large z

For the point-like charge distribution,

$$f_{\gamma/A}^{\text{PL}}(z, b) = \frac{\alpha_{\text{em}} Z^2}{\pi^2} \frac{1}{z} (zm_N K_1(zm_N b))^2, \quad (1)$$

and for large arguments $K_1(x) \sim \sqrt{\pi/2x} e^{-x}$, so

$$f_{\gamma/A}^{\text{PL}}(z, b) \propto e^{-2zm_N b}. \quad (2)$$

Physically, the Lorentz-contracted field of the nucleus is a pulse, and a pulse passing at distance b only contains frequencies up to $\omega \sim \gamma/b$. High-energy photons (large z) therefore come only from the closest encounters, and the b -integrated flux at large z is exponentially sensitive to the smallest distance that is allowed.

3 The three fluxes

PL($AnAn$). Point-like flux, measured at the target centre, with a sharp cut at $b_{\text{min}} = 14.2$ fm (`photon_flux()` in `src/photon_flux.cpp`; Eq. (3) of the paper):

$$f^{\text{PL}(AnAn)}(z) = \int_{b \geq b_{\text{min}}} d^2b f_{\gamma/A}^{\text{PL}}(z, b). \quad (3)$$

AnAn, single b (the paper’s definition).

$$f^{(AnAn)}(z) = \int d^2b f_{\gamma/A}(z, b) \Gamma_{AA}(b), \quad (4)$$

where b is “the transverse distance between the centers of the two nuclei” (paper, Sec. 3). The Woods–Saxon form factor in $f_{\gamma/A}$ changes nothing here: for $b > R_A$ the Woods–Saxon flux equals the point-like one (the field outside a spherically symmetric charge does not depend on its internal structure), and Γ_{AA} removes everything inside that range anyway.

AnAn, effective flux (our plot, our default FLUX_MODEL=EFF).

$$f_{\text{eff}}^{(AnAn)}(z) = \frac{1}{B} \int d^2r f_{\gamma/A}(z, r) \int d^2s T_B(s) \Gamma_{AA}(|\mathbf{r} - \mathbf{s}|), \quad (5)$$

where $\mathbf{r}$ is the photon’s position relative to the emitter, $\mathbf{s}$ the position of a target nucleon relative to the target centre and $b = |\mathbf{r} - \mathbf{s}|$ the centre-to-centre distance. Nucleons on the near side of the target sit a few fm closer to the emitter than the target centre does.

The paper’s own comment. The paper cites exactly this refinement and chooses not to use it: “One can in principle refine [their single- b *AnAn* formula] by taking into account details of the transverse-plane geometry of UPC events ... While it somewhat increases $f_{\gamma/A}(z)$ for large z , the photon flux is suppressed by several orders of magnitude in this region.”

4 Why their PL flux is always larger

Both of their curves measure the photon at the target centre; they differ only in the cut at small b :

b [fm]	≤ 13.8	14.2	15.5	$\gtrsim 19$
sharp cut (PL)	0	1	1	1
$\Gamma_{AA}(b)$ (inputs/Gamma_AA.dat)	0	≈ 0.01	≈ 0.5	≈ 1

Γ_{AA} removes strictly more than the sharp cut, so the single- b *AnAn* flux is below PL(*AnAn*) at every z . The gap grows with z because large- z photons come from the smallest b , which is exactly where Γ_{AA} suppresses.

5 Why our effective flux crosses above PL at large z

5.1 One exponential, evaluated at two different distances

The $e^{-2zm_N b}$ of Sec. 2 and the $e^{-2zm_N r}$ used below are the *same* function. Writing $\kappa \equiv 2zm_N$,

$$f_{\gamma/A}(z, d) \propto e^{-\kappa d}, \quad (6)$$

where d is the distance from the centre of the emitting nucleus to the point where the photon is absorbed. The two treatments only differ in what they use for d :

- single b (PL(*AnAn*) and the paper’s *AnAn*): the photon is absorbed at the target centre, $d = b$;

- effective flux: the photon is absorbed by a nucleon at position $\mathbf{s}$ inside the target, $d = r = |\mathbf{b} + \mathbf{s}|$.

Equation (6) uses the large-argument form of K_1 , valid for $zm_N d \gtrsim 1$; with $d \approx 14$ fm that means $z \gtrsim 0.015$. The effective lengths quoted below are only meaningful in that range.

5.2 The loss from Γ_{AA} : a fixed length

Replacing the sharp cut at 14.2 fm by $\Gamma_{AA}(b)$ (which rises from 0 at 14 fm to 0.5 at 15.5 fm and 1 at ~ 19 fm) effectively moves the closest allowed distance outward by some δ_{loss} :

$$L(z) \equiv \frac{f_{\text{single b}}^{(AnAn)}}{f_{\text{PL}(AnAn)}} \approx e^{-\kappa \delta_{\text{loss}}}, \quad \delta_{\text{loss}} = -\frac{\ln L}{\kappa}. \quad (7)$$

δ_{loss} is set by the width of the Γ_{AA} edge and is roughly independent of z : about 1.2–1.5 fm (table below).

5.3 The gain from averaging over nucleons: a growing length

For $b \gg |\mathbf{s}|$, a nucleon at longitudinal offset $s_{\parallel}$ towards the emitter sits at $r \approx b - s_{\parallel}$, so its flux is enhanced by $e^{\kappa s_{\parallel}}$ relative to the target centre. Averaging over the target thickness T_B ,

$$G(z) \equiv \frac{f_{\text{eff}}^{(AnAn)}}{f_{\text{single b}}^{(AnAn)}} \approx \langle e^{\kappa s_{\parallel}} \rangle_{T_B} \equiv e^{\kappa \delta_{\text{gain}}(z)}. \quad (8)$$

Two properties follow:

- $G \geq 1$ always. The target is symmetric, $\langle s_{\parallel} \rangle = 0$, but the exponential is convex, so the near-side nucleons gain more than the far-side ones lose: $\langle e^{\kappa s_{\parallel}} \rangle \geq e^{\kappa \langle s_{\parallel} \rangle} = 1$ (Jensen's inequality).
- δ_{gain} grows with z . For small κ , $\delta_{\text{gain}} \approx \kappa \langle s_{\parallel}^2 \rangle / 2 \rightarrow 0$: long-range photons see the whole target and the offsets average out. For large κ the average is dominated by the nucleons closest to the emitter, and δ_{gain} tends to the depth of the target's near edge, a few fm.

5.4 Putting them together

The ratio plotted in `flux_comparison.py` is the product of the two:

$$\frac{f_{\text{eff}}^{(AnAn)}}{f_{\text{PL}(AnAn)}} = L(z) G(z) \approx e^{\kappa (\delta_{\text{gain}}(z) - \delta_{\text{loss}})}. \quad (9)$$

Both factors are exponential in z with the same κ ; what differs is the length multiplying it. δ_{loss} is fixed, δ_{gain} grows. The effective flux therefore sits below PL while $\delta_{\text{gain}} < \delta_{\text{loss}}$ and crosses above it once δ_{gain} overtakes, at $z \approx 0.02$.

The single- b scan isolates the two factors cleanly: it has the same Γ_{AA} as the effective flux but no nucleon average. From our scans:

z	L	G	LG	δ_{loss} [fm]	δ_{gain} [fm]	$\delta_{\text{gain}} - \delta_{\text{loss}}$ [fm]
0.001	0.958	1.019	0.977	–	–	–
0.01	0.848	1.115	0.945	–	–	–
0.02	0.747	1.323	0.988	1.53	1.47	−0.06
0.03	0.660	1.690	1.115	1.46	1.84	0.38
0.05	0.522	3.390	1.770	1.37	2.57	1.20
0.07	0.420	8.196	3.442	1.30	3.16	1.86
0.1	0.312	45.07	14.04	1.23	4.01	2.78

(Lengths are omitted for $z < 0.015$, where Eq. (6) does not apply.) The loss factor alone reproduces the paper’s picture (PL always on top). The crossover in our plot comes entirely from G , whose effective length grows from ~ 1.5 fm to ~ 4 fm over this range.

6 Numbers from our code

Values of dN/dz from `out/flux_scan.csv` (PL($AnAn$) and single- b $AnAn$) and `out/flux_scan_eff_AnAn.csv` (effective $AnAn$), Pb–Pb at $\sqrt{s_{NN}} = 5.36$ TeV. Values between scan grid points are obtained by interpolating each flux’s ratio to the analytic PL formula, not the flux itself (log-log interpolation of the steeply falling flux on the coarse grid underestimates it at large z):

z	PL($AnAn$)	$AnAn$, single b (= paper)	$AnAn$, effective (= our plot)	effective/PL
0.001	7.25×10^4	6.95×10^4	7.08×10^4	0.98
0.003	1.32×10^4	1.23×10^4	1.27×10^4	0.96
0.01	1.07×10^3	9.07×10^2	1.01×10^3	0.95
0.02	1.16×10^2	8.68×10^1	1.15×10^2	0.99
0.03	18.5	12.2	20.6	1.12
0.05	0.685	0.358	1.21	1.77
0.07	3.15×10^{-2}	1.32×10^{-2}	0.108	3.4
0.1	3.71×10^{-4}	1.16×10^{-4}	5.21×10^{-3}	14.0

- The single- b column is below PL($AnAn$) everywhere, reproducing the paper’s figure: close to PL for $z < 0.01$, separating at large z .
- The effective flux is a few percent below PL for $z \lesssim 0.02$ (the stricter Γ_{AA} wins; long-range photons do not resolve the target’s size) and grows exponentially above it. A factor 14 at $z = 0.1$ corresponds to an effective closest distance about 2.8 fm shorter, from $\ln 14 \approx 2zm_N\Delta b$ (Sec. 5).

7 Consequences for our D^0 calculation

- The smallest photon fraction probed at a given (p_T, y) is $z_{\min} = m_T e^y / \sqrt{s_{NN}}$: for $p_T = 2$ GeV, $z_{\min} \approx 0.004$ at $y = 2$ and ≈ 0.03 at $y = 4$. Most of our kinematics lie where the fluxes agree to a few percent; a Pb–Pb test point at $p_T = 2$ GeV, $y = 0$ gave a $\sim 2\%$ difference between EFF and PL.

- The default DO setting (FLUX_MODEL=EFF) is *not* the paper's flux. To reproduce their treatment, run with FLUX_MODEL=WS (identical to PL in Pb–Pb), which computes $\int d^2b f_{\gamma/A}(z, b) \Gamma_{AA}(b) P_{\text{no-EM}}(b)$, their $An0n$ formula.
- Differences between the two treatments show up only at large z , i.e. forward rapidity and high p_T .